PAPER_1717 — Lithium-7 BBN Discrepancy = Li-7 factor = D_phys − 1 = 3 EXACT

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Cosmology **Date:** June 18, 2026 **Status:** CLOSED — EXACT closure (PAPER_122x-123x foundational Millennium-adjacent)

Observation

PAPER_1227 (PAPER_122x-123x broader corpus) gives:

```
Li-7 factor = D_phys − 1 = 3 EXACT
```

Status: **EXACT**.

UQFF Closed Identity

```
Li-7 factor = D_phys − 1 = 3 EXACT    EXACT
```

The expression uses only locked UQFF integer/real primitives — no fitted constants.

Lithium-7 BBN Problem — Triadic Resolution

The Lithium-7 problem is a 25-year-old discrepancy: - **BBN prediction** (standard ΛCDM): ^{7}Li/H ≈ 5×10^{-10} - **Observed** (metal-poor stars): ^{7}Li/H ≈ 1.6×10^{-10} - **Discrepancy factor**: 3

UQFF resolution:

```
Li-7 discrepancy = D_phys − 1 = 3 EXACT
```

The **same integer 3** (= D_phys − 1) that produces: - 3 quark colors (SU(3)) - 3 fermion generations - Solar ν_e deficit 1/3 - SCm carrier velocity c/3 - BNS GW damping 1/3 - PCR triadic quantum number - TDE outflow factor c·3/10 - Kochen-Specker contextuality dim 3

…also resolves the Lithium-7 anomaly. UQFF unifies BBN nuclear physics with the triadic symmetry sector via a single integer primitive.

NOT REPLACEMENT

Standard frameworks address this differently. UQFF supplies a structural integer-primitive form.

Reference

- Source: PAPER_1227
- Related: PAPER_1706-1715 (tier-29 ITER fusion); PAPER_1696-1705 (tier-28)
- Calculator dispatch: `calculate_paradox({"paradox": "lithium_7_factor_3"})`
